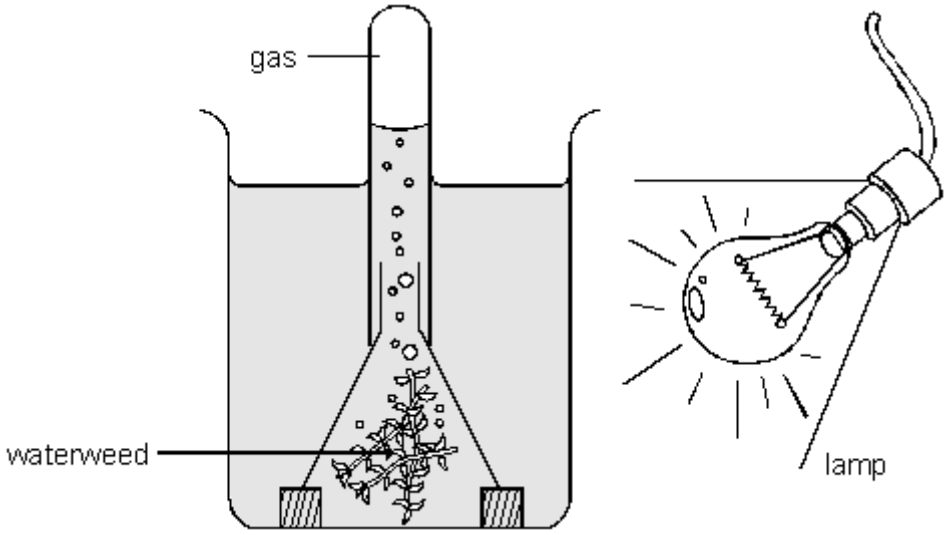
Q1. The rate of photosynthesis can be measured as the volume of oxygen released by a plant per minute. The number of bubbles of gas released per minute from waterweed is counted.



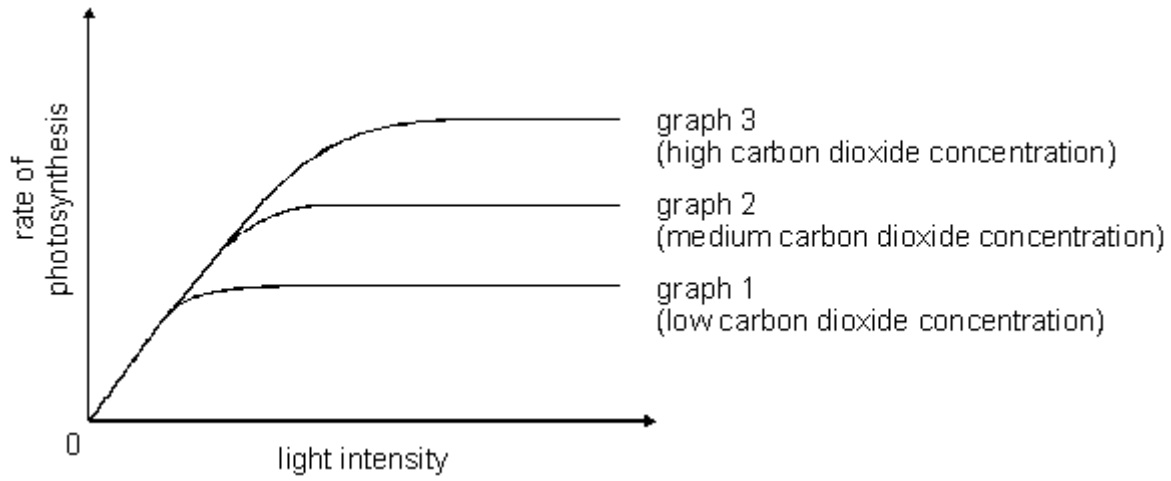
(a) The gas collected in this experiment is **not** pure oxygen. Suggest a reason for this.

.....

.....

1 mark

(b) The graphs below show the effect of light intensity on the rate of photosynthesis at different carbon dioxide concentrations.



(i) Explain what the graphs show.

.....

.....

.....

2 marks

(ii) Many years ago, gardeners used to keep a large tub of fresh manure in their greenhouses. The manure was **not** added to the soil in which the plants were growing, but the plants grew more quickly in greenhouses containing a tub of manure.

Suggest how the activity of bacteria in the manure led to an increase in the rate of photosynthesis in these plants.

.....

.....

2 marks

(c) Enzymes are involved in the process of photosynthesis.
If the temperature is too high, the enzymes stop working and photosynthesis stops.

Why do the enzymes stop working at high temperatures?

.....

.....

1 mark
Maximum 6 marks

Q1.

(a) any **one** from

- the waterweed also respires **or** produces carbon dioxide
- other gases come out of solution
  *accept 'nitrogen **or** air is dissolved in the water'*
- it contains water vapour
  *accept 'nitrogen **or** air is dissolved in the water'*

1

(b) (i) **The first mark is for the relationship between photosynthesis and light intensity.**
The second mark is for the way in which carbon dioxide affects this relationship.

any **one** from

- the rate of photosynthesis increases as light intensity increases
  accept 'photosynthesis goes up when the light gets brighter'
- at high light intensities, rate of photosynthesis does not depend on light intensity
  *do **not** accept 'the rate of photosynthesis depends on the light intensity'*

1

any **one** from

- the rate of photosynthesis cannot increase unless there is enough carbon dioxide
  accept 'photosynthesis can't go up without enough carbon dioxide'
- when there is a high light intensity, the rate of photosynthesis depends on the concentration of carbon dioxide
  accept 'at high light intensities, photosynthesis depends on the amount of carbon dioxide'
  accept 'carbon dioxide is a limiting factor'
  accept 'the rate of photosynthesis increases as carbon dioxide concentration increases'
  *do **not** accept 'photosynthesis cannot happen without carbon dioxide'*
  ***or** 'light will only increase photosynthesis up to a point'*

1

(ii) **Answers may be in either order**

- bacteria respired

1

- carbon dioxide concentration increased

  accept 'the amount of $C0_2$ went up'
  or *'it produced carbon dioxide'*
  accept 'it raises the temperature of the greenhouse'

1

(c) they are denatured

accept 'they are destroyed'
*do **not** accept 'they are killed'*

1

[6]